Homework 3

Efficient Frontier

The Excel file contains the historical price data of DJIA components collected from CRSP (Center for Research in Security Prices).

(1) (20%) Estimate the expected returns and the covariance matrix of the monthly returns. Please use the data ranging from January 2021 to December 2025. The expected return can be estimated by the arithmetic average of monthly returns.

(2) (50%) Determine the mean-variance efficient frontier. Describe the objective function, variables, and constraints of the following conditions:

  (a) (25%) No riskless asset

  (b) (25%) With riskless asset (Capital Allocation Line). Assume that R_f =3.74%. (The monthly return of R_f is therefore 3.74%/12)

  Please plot your results on the expected return standard deviation space and in **ONE** figure for comparison purposes.

(3) (15%) Find the minimum-variance portfolio (MVP) constructed by only risky assets and its expected return and standard deviation. Please report the composition (weights) of the MVP and plot the MVP in the figure of Problem (2).

(4) (15%) Assume that R_f = 3.74%. Find the tangency portfolio and its expected return and standard deviation. Please report the composition (weights) of the tangency portfolio and plot the tangency portfolio in the figure of Problem (2).

<u>Matlab function and syntax:</u>

1. [num,txt,raw] = xlsread(filename)
2. reshape(A,sz1,...,szN): Rearrange existing elements in the input data
3. mean(): Average or mean value of arrays
4. cov(): Covariance matrix
5. '   : Matrix transpose
6. sqrt(): Square root
7. zeros(): To create an array of all zeros. e.g. w=zeros(m,n)
8. length(X): To return the length of vector X
9. for loop

    e.g.

```
for j=1:10
    statement
end
```

10. To plot efficient frontier or a portfolio, you may use Matlab function plot().

    e.g.

```
plot(P_Stdev, P_Mu,'-o');
```

where P_Stdev is a vector of portfolio standard deviations and P_Mu is a vector of their corresponding mean returns. '-o' is the LineSpec option that specifies the line type, marker symbol and color: LineStyle is Solid line ('-') and marker type is circle ('o').

To plot the results of (2)(a), (2)(b), (3), and (4) in one figure, you can use “hold on” to retain the current graph when adding new graphs to it.

<u>Your computer program is required and is part of this assignment.</u>

You have to submit your results and programs by new e3.